

Neo Mohsenvand

AI/ML Research Engineer and Product Designer

Representation learning / Human-AI interaction / Agentic systems / AI-first product design

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Research engineer and product designer working across machine learning, neuroscience and human-computer interaction. I turn mathematical ideas into working systems: self-supervised EEG models, wearable memory interfaces, GPU artificial-life experiments and agentic products. MIT PhD; Oxford MSc in mathematical modelling. Previously Head of Research at BrainCo and ML research intern at Apple; now at Caterpillar.

TECHNICAL SKILLS

	Programming	Python, TypeScript, MATLAB, Mathematica, GLSL, WGSL
	Data science	Time-series analysis, biosignal processing, feature extraction, scientific visualization
	Machine learning	PyTorch, JAX, transformers, self-supervised learning, mechanistic interpretability
	Training & alignment	FSDP, DeepSpeed, Ray; SFT, PEFT, LoRA, PPO / DPO, quantization
	Evaluation	MLflow, DeepEval, benchmark design, automated and human evaluation
	Agentic engineering	LangGraph, agent harnesses, RAG, tool orchestration, human-in-the-loop workflows
	Realtime AI	OpenAI Realtime, ElevenLabs, Deepgram; low-latency multithreaded harnesses
	Inference & serving	Ollama, Transformers.js, WebGPU, ONNX / WASM, vLLM
	Cloud & data	Azure, AWS, Docker, progressive delivery, Snowflake, multimodal data pipelines
	Design & UX	Adobe Creative Cloud, UX study design, Design Systems
	Applications & UI	SvelteKit, React, Three.js / WebGL

PROFESSIONAL EXPERIENCE

 **Caterpillar Inc.** | Apr 2024 - Present

Chief AI Architect, Global Finance | Jan 2026 - Present

- **Tempo:** Designed and built a conversational investigation system with a meeting-like interview interface, combining stakeholder discovery, scheduling and OpenAI Realtime voice interviews with structured forms. Available to hundreds of internal users for idea intake, requirements gathering and conflict resolution.
- **Aperture:** Designed and built a custom agentic image-generation harness that expands terse prompts and hundreds of references into slide decks and visual assets, reducing reliance on third-party subscriptions.
- **Touchless Invoice Processing agent:** Co-developed end-to-end orchestration from email ingestion and scanned-invoice OCR through exception handling and Snowflake writes, with a detailed multi-level evaluation framework. Reduced invoices needing human review from **100% to 2%**.
- **Teaching:** Created 10+ workshop apps, including agentic RAG, talking dashboards and a voice-only data-science interface I designed. Teach engineers and senior leaders in the workshop's fifth iteration.
- Developing shared infrastructure for AI application prototyping and deployment across Global Finance.

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PROFESSIONAL EXPERIENCE / CONTINUED

Caterpillar Inc. | Apr 2024 - Present

Senior Principal Data Scientist, Global Finance | May 2025 - Jan 2026

- **Penman**: Built and deployed an enterprise financial-analysis system using **SvelteKit and Azure** to analyze earnings-call transcripts and regulatory filings for finance leaders.
- Redesigning earnings-call transcript proofreading in Penman to help finance teams identify disclosure risks.
- Built **MLflow / DeepEval** evaluation infrastructure and coordinated organization-wide benchmark development.
- Built an AI-idea prioritization agent with RICE-derived scoring, internal impact metrics and human review.
- Designed adversarial LLM analysis of draft financial disclosures.

Senior Engineering Fellow, Artificial Intelligence | Apr 2024 - May 2025

- Founding technical hire of CAT Tech AI; led four engineers delivering **12+ applications** for **8,000+ technical users** through engineering-analysis automation.
- Created **Magic Table** for million-row spreadsheet/database workflows, combining formulas and iterative local-LLM prompts to reduce cloud compute costs; developed versions with Ollama.
- Built a **ModernBERT-based embedding model** for Caterpillar products and parts; orchestrated data collection, developed a synthetic training-pair generation app, and trained on an **on-premises H200 cluster**.
- Led distributed training, data curation, SFT/PEFT, PPO/DPO alignment, and automated and human evaluation.
- Established an AI curriculum and reading group across 10+ teams, training 300+ engineers.

RESEARCH & ENGINEERING EXPERIENCE

BrainCo

Head of Research | Nov 2021 - Mar 2024

- Led **11 researchers** developing wearable AI and brain-computer interfaces; named inventor on **five provisional patent applications**.
- Led conditional-GAN reconstruction of breathing patterns from **25 Hz wearable PPG**: one-second breathing-curve feedback and **1.47 breaths/min MAE** across ten test subjects.
- Led soft-speech recognition research achieving **35% word error rate** with contact microphones versus **99-100%** with MEMS microphones in tested music/noise conditions across eleven participants.

MIT Media Lab

Postdoctoral Research Scientist (part-time) | Nov 2021 - Jun 2022

- Researched multimodal memory systems and physiological data, building on doctoral work in continuous-capture memory interfaces and self-supervised EEG representations.

Apple

Machine Learning Research Intern | Jun - Sep 2021

- Developed **parameter-efficient transformers** for biosignal event detection and continuous time-series processing.
- Built methods to inspect attention heads and activation patterns in continuous time-series models.

cortico **Cortico** | Information Design & Data Visualization Intern | Summer 2017

Built a browser WebGL engine for dynamic social-network graphs.



University of British Columbia | Data Scientist & Researcher | Sep 2013 - Aug 2014

Developed a Python library to retrieve, compose and visualize Allen Brain Atlas connectivity data in 3D.

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EDUCATION



Massachusetts Institute of Technology | PhD, Media Arts and Sciences

Completed 2021; MIT degree record: Feb 2022

Advisor: Pattie Maes. Co-advisors: Tomaso Poggio and Ed Boyden.

Thesis: Classifying and Displaying Brain-Waves through Self-Supervised Learning.

Doctoral research: first author of SeqCLR (2020), contrastive EEG learning evaluated on emotion recognition, sleep staging and abnormal-EEG detection.



University of Oxford | MSc, Mathematical Modelling and Scientific Computing | 2013

Advisor: Jared Tanner. Co-advisor: Mason Porter.

Thesis: Sparse-Signal Recovery and Compressive Sensing for Optical Functional Brain Imaging.



Tehran Polytechnic / Amirkabir University of Technology

BSc, Biomedical Engineering; BSc, Electrical Engineering | 2011

President, IEEE Student Branch.

HONOURS & AWARDS

NTT Data Fellowship | MIT, for Rhizome, a smart memory book (2020).

First place and Koch AI Prize | MIT Grand Hack, for Memoroom, VR for Alzheimer's (2017).

Bill Mitchell Design Award | MIT Media Lab (2014).

Khwarizmi Prize | First in mathematics and physics; third the previous year (2006).

Semifinalist | National mathematics and physics olympiads (2004).

INDEPENDENT RESEARCH & CREATIVE ENGINEERING



Mindview | Ternary model fine-tuning, WebGPU inference and live model visualization; Hugging Face releases.



OP-XY Agent | Playable synth replica; teaching/programming agent harness with USB MIDI hardware control.



Zilion | Parallel WebGPU Z80 emulation for artificial life and genetic programming.



jaxverse | Interactive deep-learning book with models that train live in the browser.



Embedding Playground | Hands-on retrieval, classification and clustering experiments with browser inference.



Voicebook | Voice document reader: AI scroll/highlight; local ONNX inference or optional cloud models.



harnessXray | Agent observability: streamed request capture, record/replay and persistent checkpoints.



GradientLab | Interactive optimization lab: optimizer races, curvature analysis and lessons; 5M lifetime visitors.

SELECTED PUBLICATIONS

Contrastive Representation Learning for Electroencephalogram Classification

First author. ML4H NeurIPS Workshop, PMLR 136 (2020).

Soft Speech, Loud World: Bone Conduction Microphones Enhance Voice Assistant Interaction

Principal investigator. IEEE International Conference on Consumer Electronics (2024).

Generating Breathing Patterns in Real-Time: Low-Latency Respiratory Phase Tracking From 25 Hz PPG

Principal investigator. HealthyIoT / HealthWear, Springer LNICST 530 (2024).

Optical-flow analysis toolbox for characterization of spatiotemporal dynamics in mesoscale optical imaging of brain activity

Coauthor. NeuroImage 153, pp. 58-74 (2017).

Rumor Gauge: Predicting the Veracity of Rumors on Twitter

Coauthor. ACM Transactions on Knowledge Discovery from Data 11(4), Article 50 (2017).

Spontaneous cortical activity alternates between motifs defined by regional axonal projections

Coauthor. Nature Neuroscience 16, pp. 1426-1435 (2013).

SELECTED MEDIA COVERAGE

MIT Technology Review | Profile of wearable memory research (2018)

Quartz | Mini-documentary: Wearable technology that records memories (2018)

Fast Company | DataVRse: immersive data exploration (2016)